Lecture 2: Random Variables and Their Distributions

Math for Data Science · 2026

Jan Gieseler

2026-09-08

Agenda

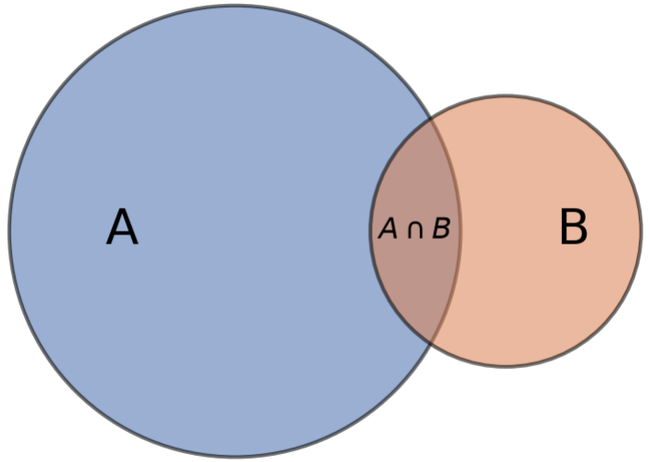
- Part I: Recap
- Part II: Bayes' Rule
- Part III: Independence
- Part IV: The Monty Hall Problem
- Part V: Binomial distribution
- Part VI: Random Variables

Part I: Recap

Events and sets

An event $A \subseteq S$ is any subset of the sample space.

Operation	Symbol	Meaning	Notation
Intersection	$A \cap B$	*Both* A AND B	$P(A, B) = P(A \cap B)$
Union	$A \cup B$	A OR B occurs	$P(A \cup B) = P(A) + P(B) - P(A, B)$
Complement	A^c	A does NOT occur	$P(A^c) = 1 - P(A)$



Counting table

		order matters: yes	order matters: no
replacement	with	n^k	$\binom{n+k-1}{k}$
	without	$n(n-1)\cdots(n-k+1)$	$\binom{n}{k}$

Memorize the derivation with a story not the result, e.g.

- tossing a n sided die k times and caring about order, 12 and 21 are differnet
- picking k cards from a deck of n
- selecting a commitee of size k from a group of size n.
- tossing a n sided die k times and not caring about order, 12 and 21 are the same

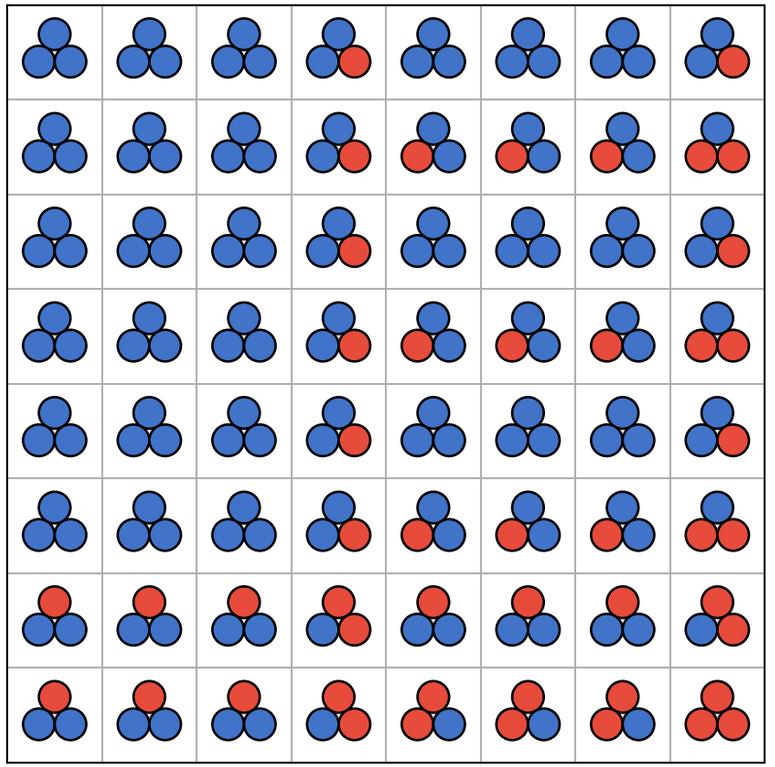
Counting events and outcomes

Say we consider events which make statements about the order at which red and blue occur. In that case the order of colors does matter for the definition of an event.

Blue	Red	Events	Outcomes per event	Outcomes
3	0	$\binom{3}{3} = 1$	$3^3 \times 1^0 = 27$	27
2	1	$\binom{3}{2} = 3$	$3^2 \times 1^1 = 9$	27
1	2	$\binom{3}{1} = 3$	$3^1 \times 1^2 = 3$	9
0	3	$\binom{3}{0} = 1$	$3^0 \times 1^3 = 1$	1
		8		**64**

However, what does not matter is the order of the blue sides when we count the events. That is why we use the binominal coefficient.

Count for yourself

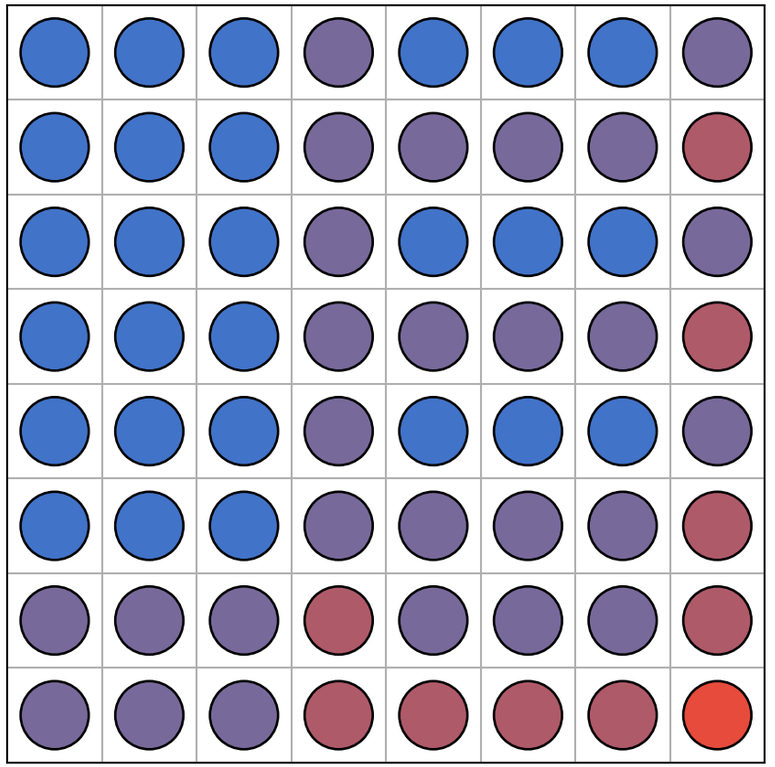


Counting events and outcomes

Say we consider events which do not make statements about the order at which red and blue occur. In that case the order of colors does not matter for the definition of an event.

Blue	Red	Events	Outcomes per event	Outcomes
3	0	1	$\binom{3}{3} \times 3^3 \times 1^0 = 1 \times 27$	27
2	1	1	$\binom{3}{2} \times 3^2 \times 1^1 = 3 \times 9$	27
1	2	1	$\binom{3}{1} \times 3^1 \times 1^2 = 3 \times 3$	9
0	3	1	$\binom{3}{0} \times 3^0 \times 1^3 = 1 \times 1$	1
		4		**64**

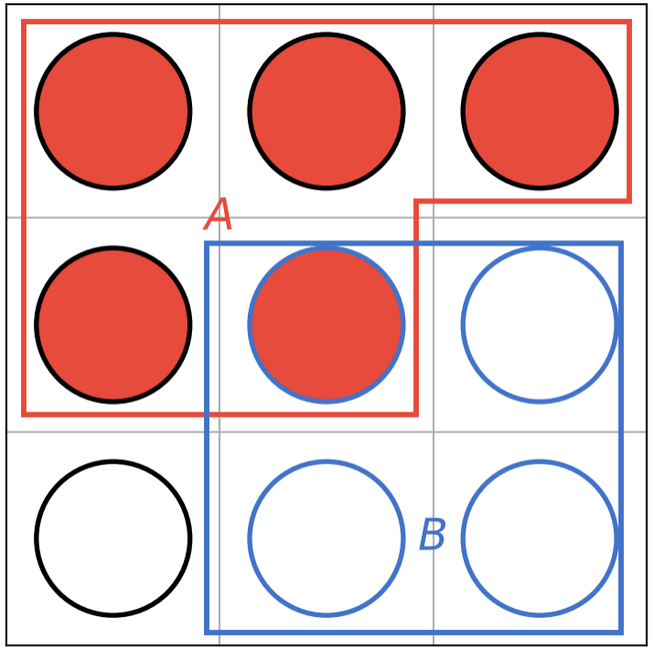
Count for yourself



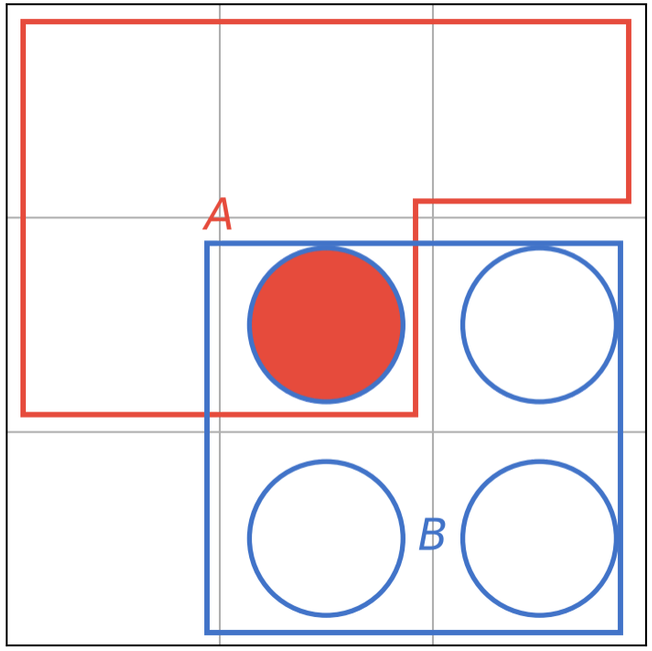
Conditional probability

update your probabilities as you obtain new information $P(A|B) = P(A \cap B)/P(B)$

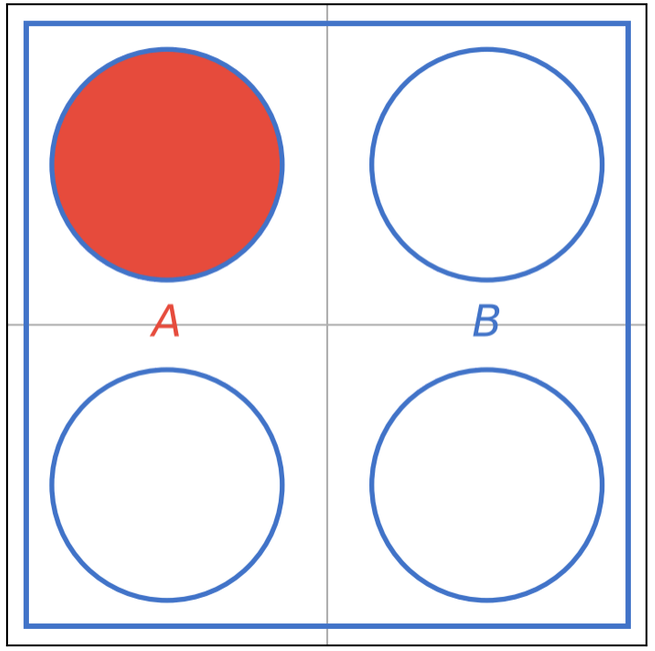
Before: all of S in play and P(A)=5/9



After conditioning on B: only B matters



B is the new S: $P(A \mid B) = \frac{1}{4}$



⚠ not symmetric in general.

$P(B \mid A) \neq P(A \mid B)$

Law of Total Probability (LOTP)

Let $A_1, \ldots, A_n$ be a partition of the sample space S, meaning that the A_i's are disjoint events and their union is S, with $P(A_i) > 0$ for all i.

Then:

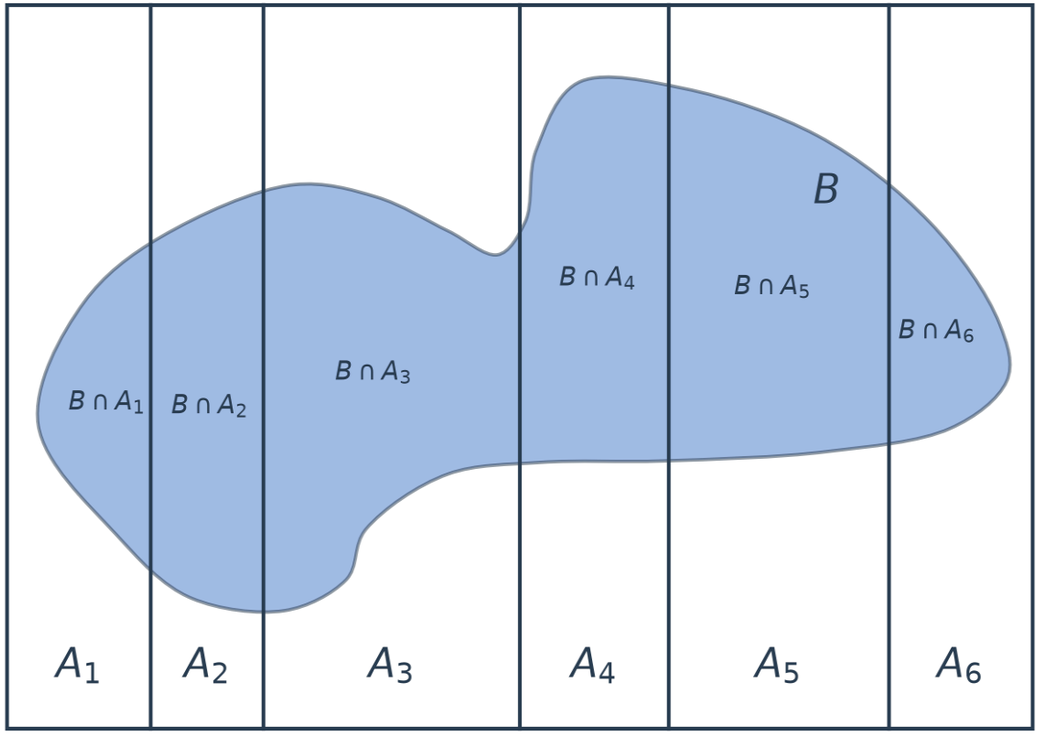
$$P(B) = \sum_{i=1}^{n} P(B \mid A_i)\, P(A_i)$$

$$= P(B \mid A_1)\, P(A_1) + \cdots + P(B \mid A_n)\, P(A_n)$$

For more intuition, work through:

- Proof of Theorem 2.3.6
- Examples 2.3.7, 2.3.8, 2.3.9, 2.3.10

This is easiest to see with a picture; see complete proof under **Theorem 2.3.6.**



After Fig. 2.3 in Blitzstein & Hwang

Part II: Bayes' Rule

Bayes' Rule

Recall the definition of conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}; \quad P(B \mid A) = \frac{P(B \cap A)}{P(A)}$$

rearanging we get

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A)$$

since $P(A \cap B) = P(B \cap A)$, we get

Bayes' rule converts $P(B \mid A)$ (easy to model) into $P(A \mid B)$ (what we actually want).

$$\boxed{P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}}$$

Disease Testing

We often use **Bayes' Rule** and the **LOTP** together. For instance, consider disease testing.

- Let D be the event that you have a disease, and suppose 1% of the population has it at this moment, i.e. $P(D) = 0.01$.
- Let T be the event that you test positive.
- Suppose the test is 95% accurate, by which we mean:
  - $P(T \mid D) = 0.95$ (*sensitivity* - true positive rate)
  - $P(T^c \mid D^c) = 0.95$ (*specificity* - true negative rate)

Suppose you test positive for the disease. What is the probability you really are infected, given the positive test result, i.e. what is $P(D \mid T)$?

Disease Testing: Bayes + LOTP

Applying Bayes' Rule and the LOTP, we have:

$$P(D \mid T) = \frac{P(T \mid D)\,P(D)}{P(T \mid D)\,P(D) + P(T \mid D^c)\,P(D^c)}$$

$$= \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.05 \times 0.99} \approx \mathbf{0.16}$$

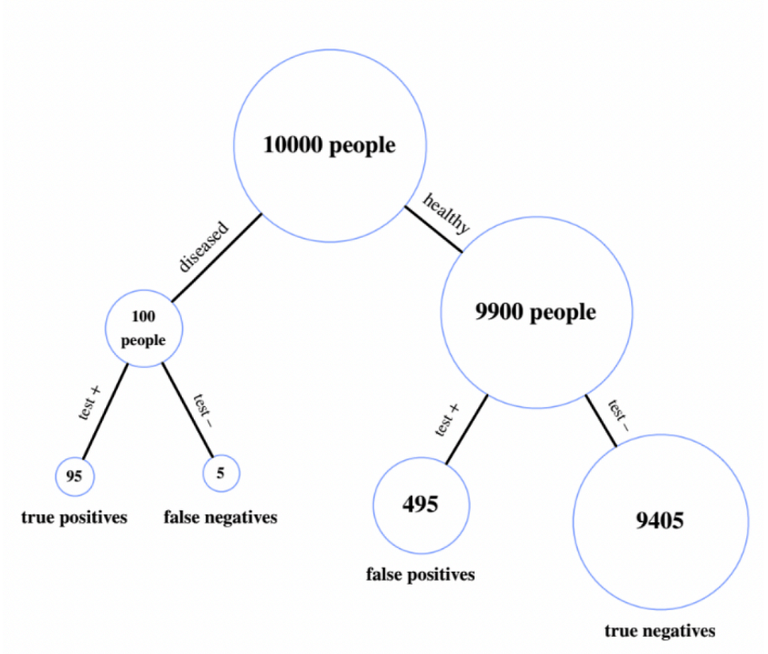
⚠ **rarity of disease**

Despite a 95%-accurate test, a positive result only means a **16% chance** of actually having the disease. Why?

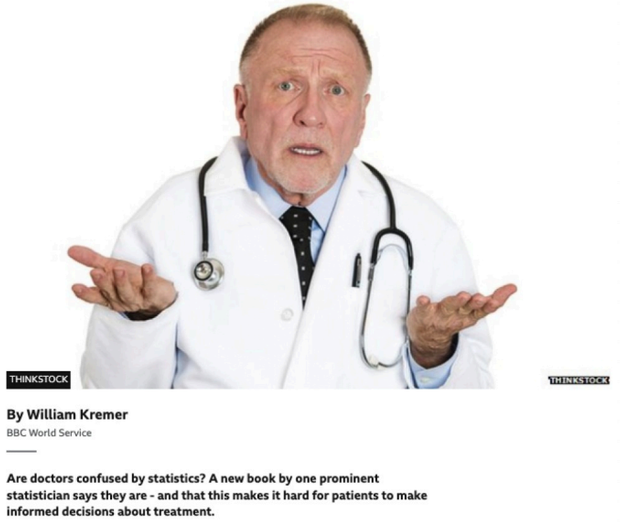
The rarity of the disease $P(D) = 1\%$ overwhelms the test accuracy.

Think in Natural Frequencies

Imagine testing **10,000 people** - the same numbers, as counts instead of probabilities:



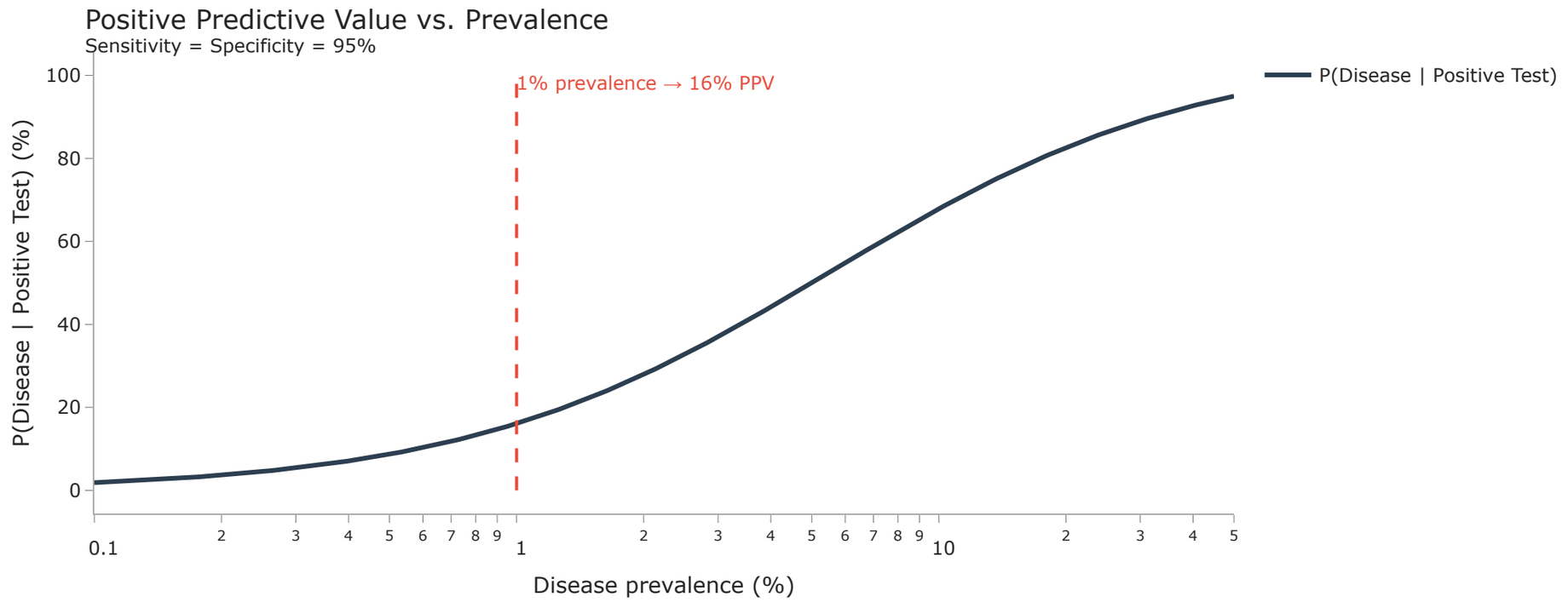
Do doctors understand this?



Of the $95 + 495 = 590$ positive tests, only **95 truly have the disease** - about **16%**.

Are doctors confused by statistics? - bbc.com/news/magazine-28166019

The Base Rate Fallacy



Extra Conditioning

Everything holds when we condition on an additional event E:

Bayes' Rule: Provided that $P(A \cap E) > 0$ and $P(B \cap E) > 0$

$$P(A \mid B, E) = \frac{P(B \mid A, E)\, P(A \mid E)}{P(B \mid E)}$$

LOTP: Let $A_1, \ldots, A_n$ be a partition of S. Provided that $P(A_i \cap E) > 0$ for all i, we have

$$P(B \mid E) = \sum_{i=1}^{n} P(B \mid A_i, E)\, P(A_i \mid E)$$

> ⓘ **All the usual rules apply**
>
> This follows from the simple fact that conditional probabilities are probabilities.

For further intuition see **Examples 2.4.4 and 2.4.5**.

Part III: Independence

Independence of Events

Events A and B are independent if:

$$P(A \cap B) = P(A) \times P(B)$$

When $P(A) > 0$, $P(B) > 0$, this is equivalent to:

$$P(A \mid B) = P(A)$$

That means learning that B occurred gives no information whatsoever about A.

Independence is symmetric: if $A \perp B$ then $B \perp A$.

Independence ≠ Disjoint

Independent and disjoint events are opposite concepts - don't confuse them!

Property	Meaning	Information?
Independent	$P(A \cap B) = P(A)P(B)$	Independence means A gives you no information about B
Disjoint	$P(A \cap B) = 0$	Knowing A occurred tells you that B definitely did not occur, which is information.

Disjoint and independence

Disjoint events (with $P(A), P(B) > 0$) **cannot** be independent - knowing A happened tells you B definitely didn't.

Disjoint events can only be independent if $P(A) = 0$ or $P(B) = 0$. See also **2.5.2** in Blitzstein and Hwang.

If A and B are independent (i.e $A \perp B$), then $A \perp B^c$, $A^c \perp B$, and $A^c \perp B^c$

Proof: To show that A, B^c are independent: $P(B^c \mid A) = 1 - P(B \mid A) = 1 - P(B) = P(B^c)$

See Proposition **2.5.3** in Blitzstein and Hwang for further detail.

Independence of Three Events

A, B, C are **independent** if *all of the following four* conditions hold:

$$
\begin{aligned}
P(A \cap B) &= P(A)P(B) \\
P(A \cap C) &= P(A)P(C) \\
P(B \cap C) &= P(B)P(C) \\
P(A \cap B \cap C) &= P(A)P(B)P(C)
\end{aligned}
$$

The first three conditions are called *pairwise independence*. Pairwise independence alone does **not** imply full independence, the fourth condition is must also be satisfied.

Example: Pairwise ≠ Joint Independence

Example 2.2.5

Consider two fair, independent coin tosses and let:

- A be the event that the first toss is Heads
- B be the event that the second toss is Heads
- C be the event that the both tosses have the same result

- Are A and B pairwise independent? **Yes** - flips are independent
- Are A and C pairwise independent? **Yes** - $P(C \mid A) = 1/2 = P(C)$
- Are B and C pairwise independent? **Yes** - by symmetry
- Are A, B, C jointly independent? **No** - $P(A \cap B \cap C) = 1/4 \neq P(A)P(B)P(C) = 1/8$

Independence of Many Events

For n events $A_1, A_2, \ldots, A_n$ to be independent, we require:

- Any pair to satisfy $P(A_i \cap A_j) = P(A_i)P(A_j)$ for $i \neq j$
- Any triplet to satisfy $P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k)$ for distinct i, j, k
- And similarly all quadruplets, quintuplets, and so on…

Conditional Independence

A and B are *conditionally independent* given E if:

$$P(A \cap B \mid E) = P(A \mid E)\, P(B \mid E)$$

Some important notes about conditional independence

1. Conditional independence given E does not imply conditional independence given E^c (**Example 2.5.9**)
2. Conditional independence does not imply independence (**Example 2.5.10**)
3. Independence does not imply conditional independence (**Example 2.5.11**)

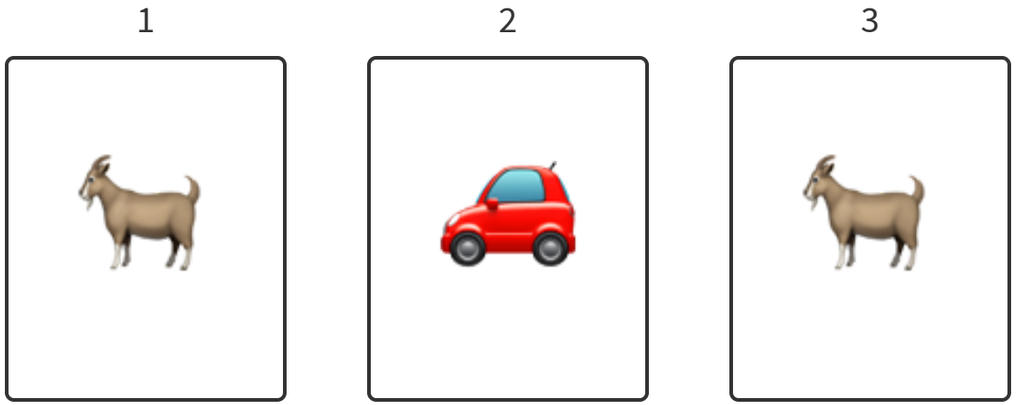
Further intuition: see **Example 2.5.12**.

Part IV: The Monty Hall Problem



The Monty Hall Problem

In this game, you are shown **three closed doors**. Behind two doors are **goats**; behind the third door is a **car**. You will get to keep whatever is behind a door of your choosing.



You pick a door. Then, the game show host opens one of the remaining doors to **reveal a goat**. *(The host always has to reveal a goat, never the car.)* Now, he gives you a chance to **switch** your guess or **stick** to your choice. **What do you do?**

Monty Hall: Setting Up the Math

Without loss of generality, suppose you pick **Door 1** and Monty opens **Door 2**. What is the probability you get a car if you stick with **Door 1**?

- Let M_2 be the event that the host opens Door 2
- Let C_i be the event that the car is behind Door i; prior $P(C_i) = \frac{1}{3}$

The key likelihoods - **host behavior depends on where the car is**:

Situation	$P(M_2 \mid \cdot)$
Car at Door 1 (you chose right)	$\frac{1}{2}$ - host picks randomly between 2 and 3
Car at Door 2 (host can't open it)	0
Car at Door 3 (host must open Door 2)	1

By LOTP: $P(M_2) = P(M_2|C_1)P(C_1) + P(M_2|C_3)P(C_3) + P(M_2|C_3)P(C_3) = \frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{1}{2}$

Monty Hall: Applying Bayes' Rule

This is a straightforward application of Bayes' Rule and the LOTP.

$$P(C_1 \mid M_2) = \frac{P(M_2 \mid C_1)\,P(C_1)}{P(M_2)}$$

From before: $P(M_2) = \frac{1}{2}, P(C_1) = \frac{1}{3}, P(M_2 \mid C_1) = \frac{1}{2}$

$$P(C_1 \mid M_2) = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}$$

$$\Rightarrow P(C_3 \mid M_2) = 1 - \frac{1}{3} = \frac{2}{3}$$

Always switch

Switching wins with probability $\frac{2}{3}$; staying wins with $\frac{1}{3}$.

Part V: Binomial distribution

General solution for two outcomes

- flippig two heads in three coin tosses
- buying one million lottery tickets and winning at least once
- rolling fewer than three 20s in 10 rolls of a 20-sided die

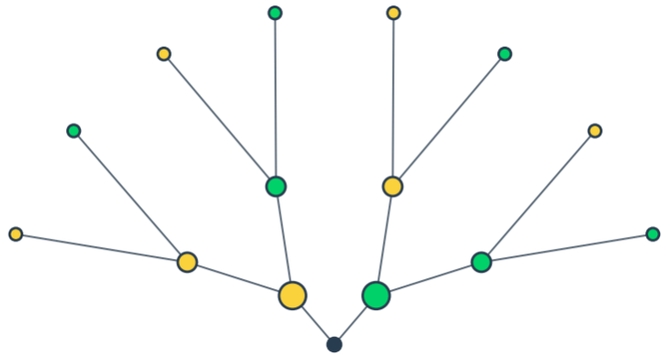
- k The number of outcomes we care about
- n The total number of trials
- p The probability of the event happening

Flippig two heads in three coin tosses

Events of interest:

[🟡, 🟡, 🟢], [🟡, 🟢, 🟡], [🟢, 🟡, 🟡]

- $k = 2$ The number of outcomes we care about
- $n = 3$ The total number of trials
- $p = 1/2$ The probability of the event happening



Multiplication rule

$$p(🟡, 🟡, 🟢) = p(🟡, 🟢, 🟡) = p(🟢, 🟡, 🟡) = \left(\frac{1}{2}\right)^2 \times (1 - \left(\frac{1}{2}\right))^1 = 0.125$$

Hence,

$$\Rightarrow p(2🟡; 3) = 3 \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^1 = 0.375$$

Binomial distribution

The general form for k desired outcomes in n tosses when the probability in a single toss is p is given by the Binomial distribution

$$\mathcal{B}(k; n, p) = \binom{n}{k} \times p^k \times (1-p)^{n-k}$$

ⓘ **Note that there is nothing more to this rule than counting and applying the rules of probability.**

Notice the similarity to the the 4 sided die example from before:

Blue	Red	Outcomes per event	Outcomes	$\mathcal{B}(k; n, p)$
3	0	$\binom{3}{3} \times 3^3 \times 1^0 = 1 \times 27$	27	$\frac{27}{64}$
2	1	$\binom{3}{2} \times 3^2 \times 1^1 = 3 \times 9$	27	$\frac{27}{64}$
1	2	$\binom{3}{1} \times 3^1 \times 1^2 = 3 \times 3$	9	$\frac{9}{64}$
0	3	$\binom{3}{0} \times 3^0 \times 1^3 = 1 \times 1$	1	$\frac{1}{64}$
			64	

Bernoulli distribution

The Bernoulli distribution is a special case of the binomial distribution with $n = 1$.

$$\text{Bern}(k; p) = p^k \times (1-p)^{1-k} \quad \text{for} \quad k \in \{0, 1\}$$

- $k = 0$: $\text{Bern}(0; p) = 1 - p$
- $k = 1$: $\text{Bern}(1; p) = p$



Jacob Bernoulli

Part VI: Random Variables

Random Variables

Definition: Given an experiment with sample space S, a random variable is a function from the sample space S to the real numbers $\mathbb{R}$.

Thus, a random variable X assigns a numerical value $X(s)$ to each possible outcome s of the experiment.

> ⓘ **The experiment makes X random; the mapping is deterministic.**

In other words: the random variable is what connects the random processes we have been learning about to data.

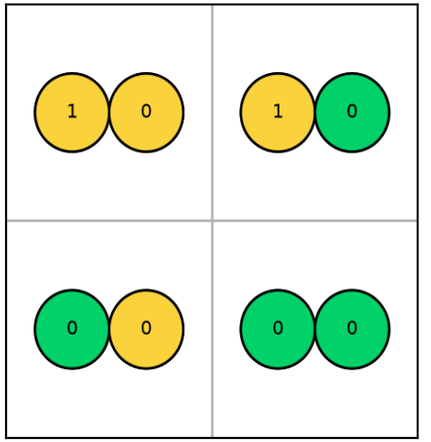
Some Examples of Random Variables

Example 3.1.2 Consider an experiment where we toss a fair coin twice. The sample space consists of four possible outcomes: $S = \{HH, HT, TH, TT\}$.

1. Let X be the number of heads. This assigns the values:
   $X(HH) = 2,\ X(HT) = 1,\ X(TH) = 1,\ X(TT) = 0$

2. Let Y be the number of tails. This assigns the values:
   $Y(HH) = 0,\ Y(HT) = 1,\ Y(TH) = 1,\ Y(TT) = 2$
   Note that $Y(s) = 2 - X(s)$ for all s.

3. Let I be an indicator for whether the first toss is heads. Then I assigns 1 to HH and HT and 0 to TH and TT.



More on Discrete Random Variables

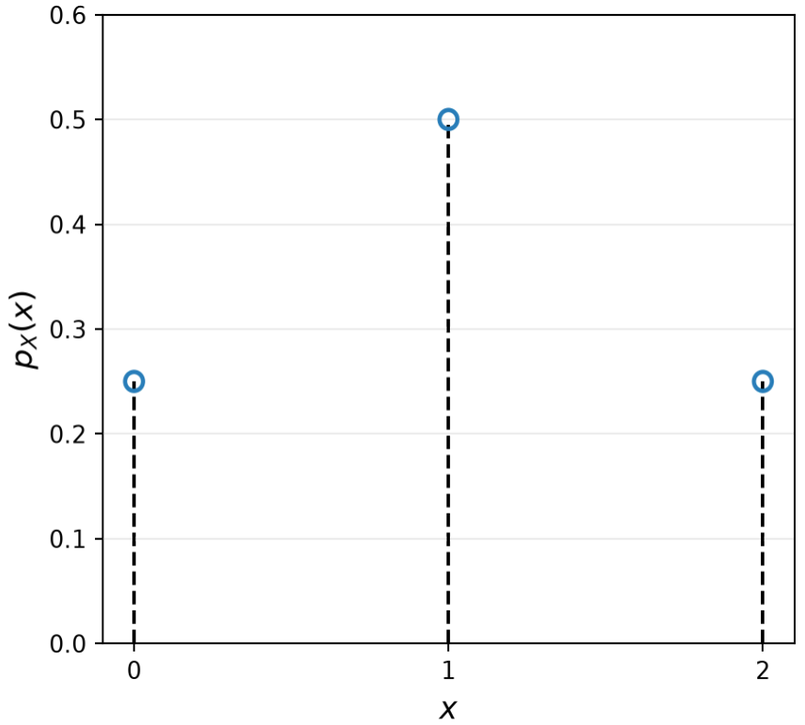
The support of X is defined as all the values x such that $P(X = x) > 0$.

The probability mass function (PMF) of X is the function $p_X(x) = P(X = x)$. This is positive if x is in the support of X, and 0 otherwise.

Returning to the coin toss example, let X again be the number of heads in two coin tosses. Again, X is 0 if TT occurs, 1 if HT or TH occurs, and 2 if HH occurs. Then the PMF of X is:

$$p_X(0) = P(X = 0) = 1/4$$
$$p_X(1) = P(X = 1) = 1/2$$
$$p_X(2) = P(X = 2) = 1/4$$

and $p_X(x) = 0$ for all other values of x.



Two Properties of a Valid PMF

The PMF p_X of X must satisfy the following two criteria:

- Nonnegative: $p_X(x) > 0$ if $x = x_j$ for some j, and $p_X(x) = 0$ otherwise;
- Sums to 1: $\sum_{j=1}^{\infty} p_X(x_j) = 1$

Notation: random variables and their distributions

A random variable is said to have the Bernoulli distribution with parameter p if $P(X = 1) = p$ and $P(X = 0) = 1 - p$, where $0 < p < 1$.

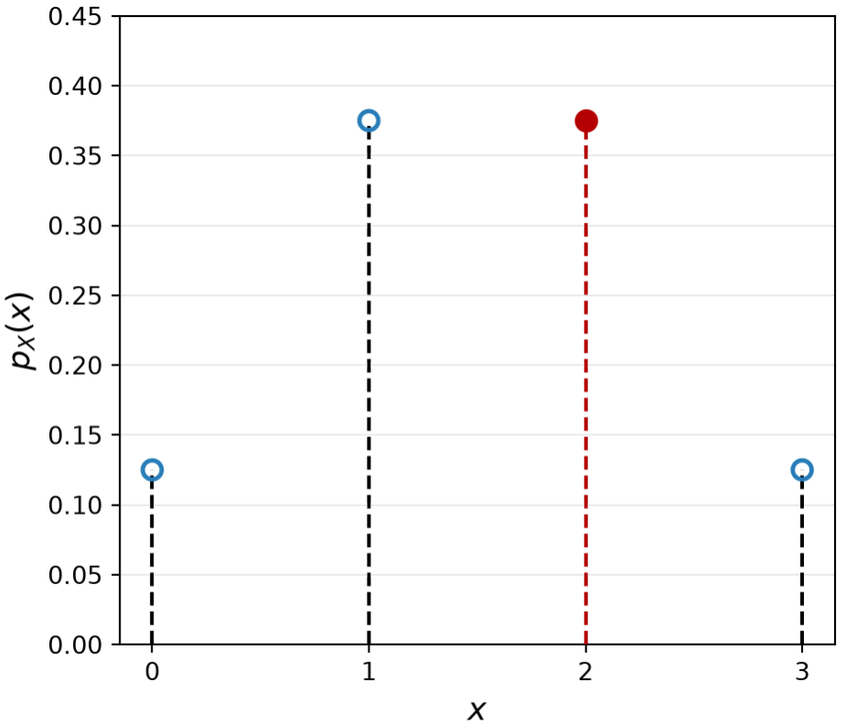
We write this as $X \sim \text{Bern}(p)$, where p is the success probability.

For a random variable X that follows a Binomial distribution with parameters n and p, where $0 < p < 1$ and $n > 1$, we write $X \sim \mathcal{B}(n, p) = \binom{n}{X} \times p^X \times (1 - p)^{n-X}$, where p is the success probability of a single trial and n the number of trials.

Some examples: coin

Getting x heads in three tosses

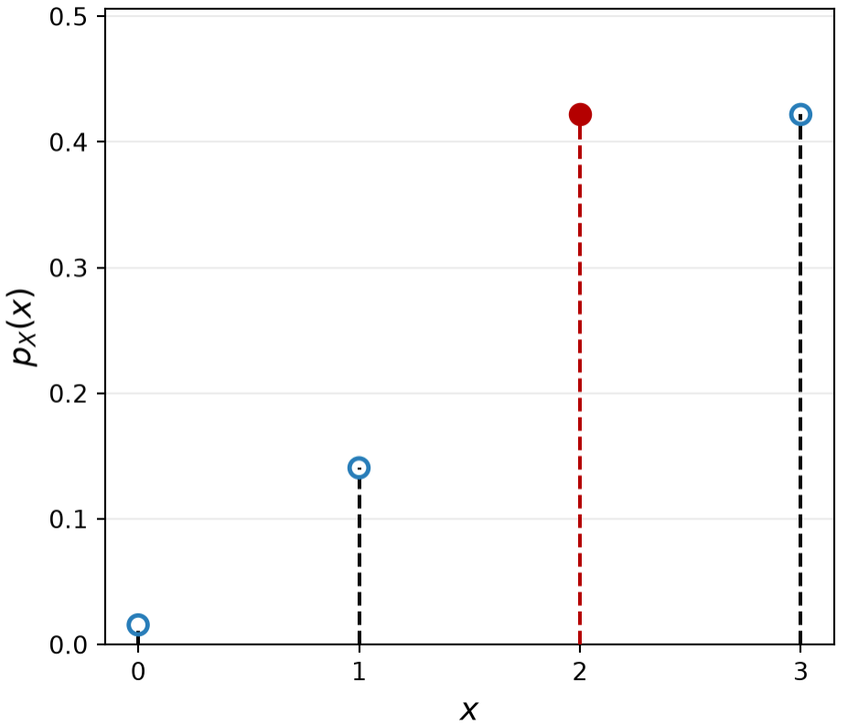
$$X \sim \mathcal{B}(3, \frac{1}{2})$$



Some examples: 4 sided die

Getting x blue in three tosses of a four sided die

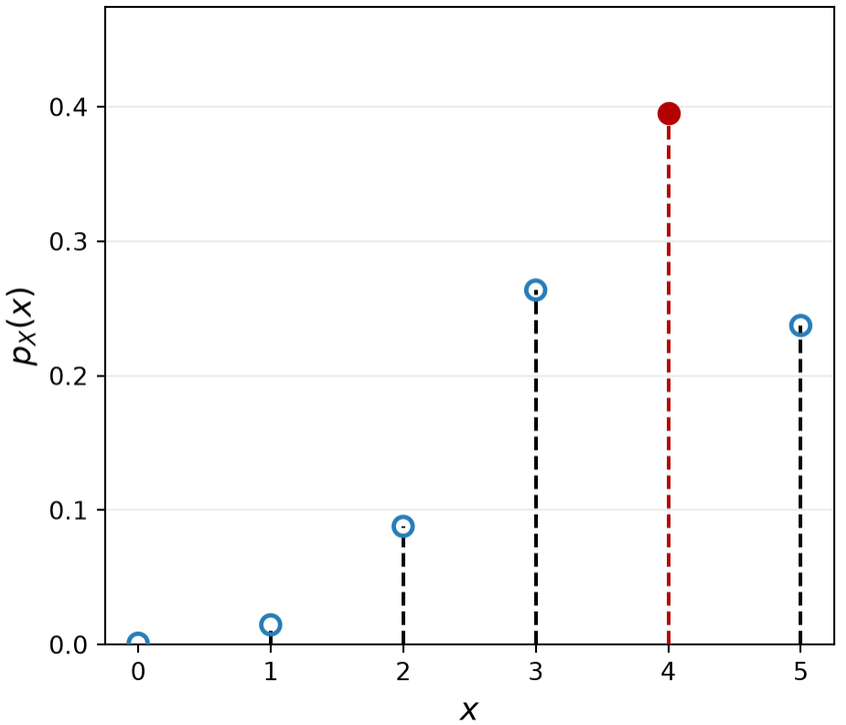
$$X \sim \mathcal{B}(3, \frac{3}{4})$$



Some examples: 4 sided die

Getting x blue in five tosses of a four sided die

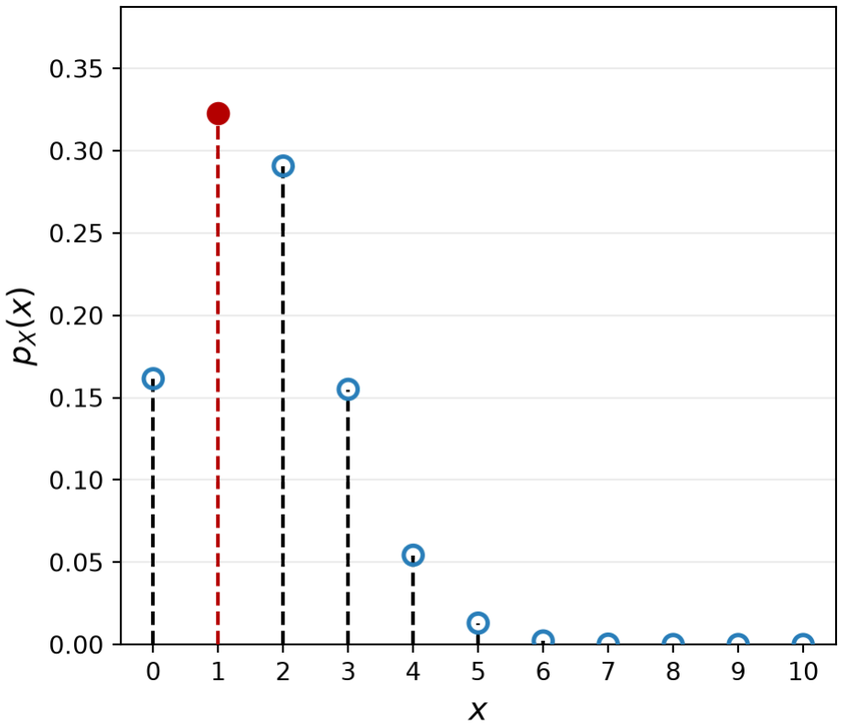
$$X \sim \mathcal{B}(5, \frac{3}{4})$$



Some examples: 6 sided die

The probability of getting a 6 x times when tossing a 6 sided die 10 times.

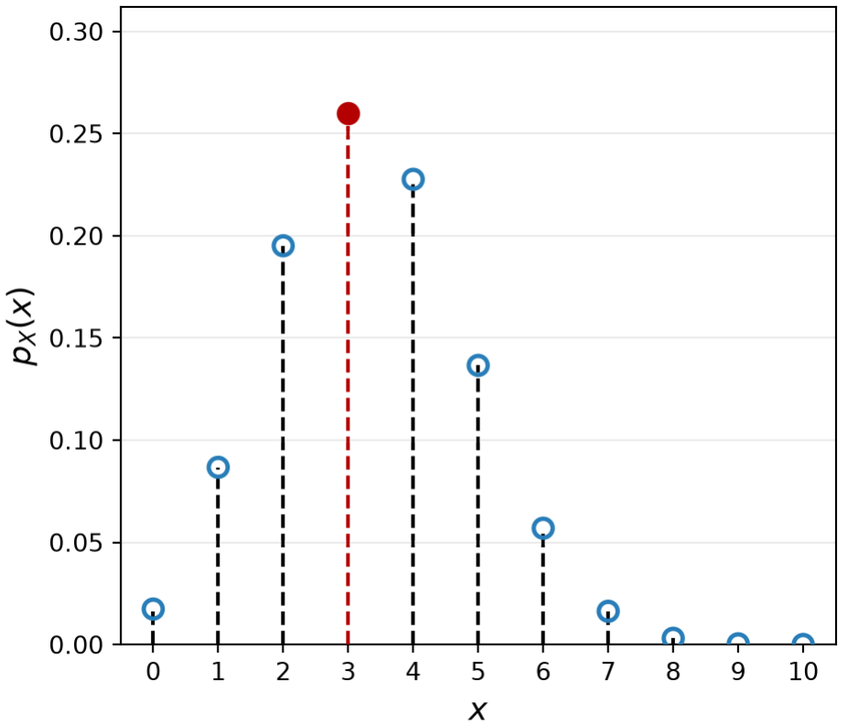
$$X \sim \mathcal{B}(10, \frac{1}{6})$$



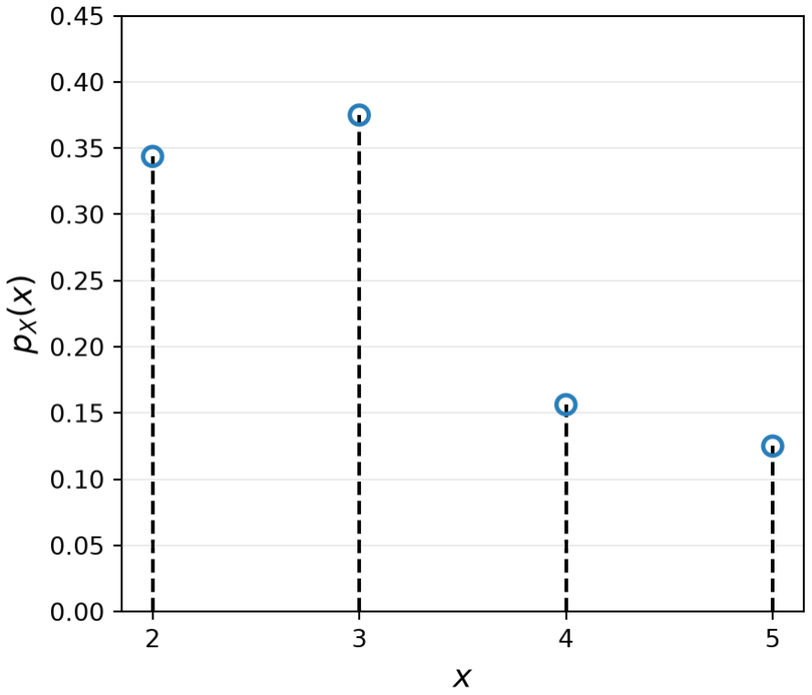
Some examples: 6 sided die

The probability of getting a 5 or a 6 x times when roling a 6 sided die 10 times.

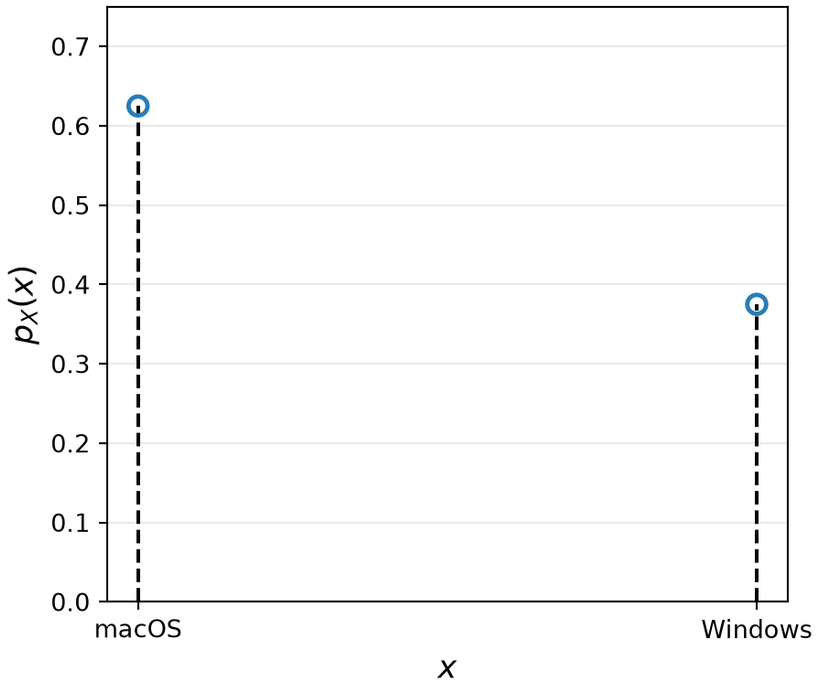
$$X \sim \mathcal{B}(10, \frac{2}{6})$$



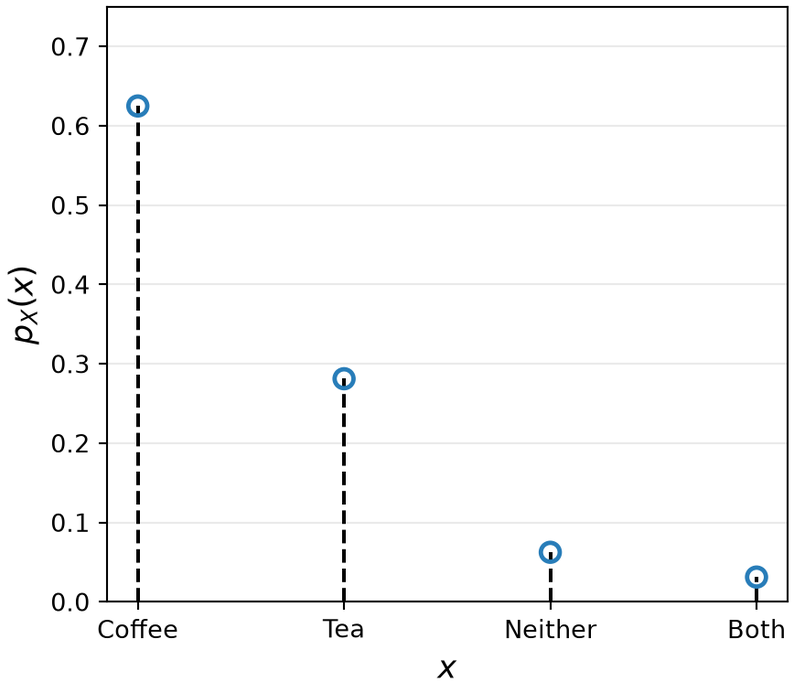
Some examples: Languages spoken by people in this class



Some examples: Laptop



Some examples: Coffee vs Tea



Discrete and Continuous Random Variables

Discrete Random Variables: A discrete random variable may take on only a countable number of distinct values such as $0, 1, 2, 3, 4, \ldots$ Discrete random variables are usually (but not necessarily) counts.

Continuous Random Variables: By contrast, continuous random variables can take on any real value in an interval.

Check your understanding: are the following random variables discrete or continuous?

- Height, weight, number of siblings, years of schooling, number of major earthquakes in the next five years, money in your bank account